MIL–A–3427A
29 MAY 1956

SUPERSEDING
MIL–S–3427
1 MARCH 1951

MILITARY SPECIFICATION

AMMETERS AND SHUNT, AUTOMOTIVE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers ammeters and one type of attaching shunt for use with the ammeters in automotive vehicular applications. The ammeters carry a small fraction of the charging or discharging current, the shunt being calibrated to such a resistance value that maximum charge or discharge will be so indicated by the ammeters.

1.2 Classification. Ammeters shall be of the following types as specified (see 6.1).

Type I—Waterproof.
Type II—Non-Waterproof.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L–L–31	—Laminated Thermosetting Sheets, Rods and Tubes.
NN–B–591	—Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment).
NN–B–621	—Boxes, Wood, Nailed and Lock-Corner.
NN–B–631	—Boxes; Wood, Wirebound (For Domestic Shipment).
QQ–M–151	—Metals; General Specification for Inspection of.
QQ–P–416	—Plating, Cadmium (Electro-Deposited).
QQ–S–561	—Solder; Silver.
QQ–S–571	—Solder; Soft (Tin, Tin-Lead, and Lead Silver).
TT–C–595	—Colors; (for) Ready-Mixed Paints.
TT–E–485	—Enamel; (for) Drums and Other Metal Products, Rust-Inhibiting.
TT–E–529	—Enamel, Synthetic, Semi-Gloss.
TT–P–636	—Primer, Paint; Synthetic (for Ferrous Metal and Wood Surfaces).
TT–R–58	—Radioactive-Luminous-Compound and Adhesives.
PPP–B–566	—Boxes, folding, Paperboard.
PPP–B–601	—Boxes, Wood, Cleated-Plywood.
PPP–B–676	—Boxes, Set-Up, Paperboard.

MILITARY

JAN–S–44	—Shock-Testing-Mechanism for Electrical-Indicating Instruments (2½- and 3½-inch, Round, Flush-Mounting, Panel-Type).
JAN–P–106	—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed
MIL–B–107	—Boxes, Wood, Wirebound (Military Type).
JAN–B–108	—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL–P–116	—Preservation, Methods of.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

JAN–P–120 —Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.
MIL–B–121 —Barrier-Materials, Greaseproof.
MIL–B–138 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined (for Weight of Contents not Exceeding 500 Pounds).
MIL–B–4229 —Boxes; Paperboard, Metal-Stayed.
MIL–L–10287 —Lacquer, Cellulose Nitrate (for Identification of Small Arms).
MIL–D–13570 —Dust, Testing by Exposure to.
MIL–E–13856 —Electrical Components; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.
MIL–F–13927 —Fungus Resistance Test; Automotive Electrical Components, General Specification for.
MIL–C–13993 —Connector, Plug, Electrical Connector, Receptacle, Electrical (Waterproof—for Military Vehicles).

STANDARDS

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.
MIL–STD–130—Identification Marking of U. S. Military Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Ammeters and shunt furnished under this specification shall be products which have been tested and have successfully passed the qualification tests specified in 4.3.1.1.

3.2 Materials. The materials shall be as specified herein, on the applicable drawings, and in the applicable specifications. When not specifically covered by this or applicable specifications, the materials shall be of good quality and shall be free of defects which may affect the serviceability and appearance of the ammeters and shunts. Materials which have a tendency to warp or absorb moisture and thereby cause change in its dielectric properties, shape, or appearance, shall not be used.

3.2.1 *Protective treatment.* All parts subject to corrosion shall be fabricated of corrosion-resistant materials or shall be suitably treated to resist corrosion. All parts subject to the growth of fungi shall be fungi-resistant or shall be suitably treated to resist the growth of fungi.

3.2.2 *Solder.* Solder employed in making electrical connections shall be a rosin-core type conforming to composition SN 60 of Specification QQ–S–571 or silver solder conforming to specification QQ–S–561, as specified.

3.3 Construction. Ammeters and shunt furnished under this specification shall be substantially constructed in the forms and to the dimensions shown on applicable drawings (see 6.1).

3.4 Ammeters.

3.4.1 *Rating.* Ammeters shall be constructed to operate on 24-volt, direct-current electrical systems.

3.4.2 *Resistance.* The internal resistance and the amperage required to produce full-scale deflection on the dials of the ammeters shall be as shown on applicable drawings (see 6.1).

3.4.3 *Mounting.* Ammeters shall be provided with a bezel, a U-shaped clamp, and the necessary lockwashers and nuts in accordance with applicable drawings for panel mounting. Studs, for holding the U-shaped

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

clamp, shall be securely anchored to the ammeter case in such a manner as to prevent their turning.

3.4.4 *Connectors and terminals.* Ammeters shall be provided with connectors or terminals as shown on applicable drawings. Connectors shall be of the type and style shown on the applicable drawings (see 6.1) and, for type I ammeters, shall conform to Specification MIL–C–13993.

3.4.5 *Cases.* Ammeter cases shall be fabricated of metal to the contour and dimensions shown on applicable drawings (see 6.1). Each case shall be provided with four elongated windows equally spaced around its circumference to light the dial face from an outside source.

3.4.6 *Dials.*

3.4.6.1 *Material.* Unless otherwise specified, the dials shall be fabricated of metallic material of a suitable thickness.

3.4.6.2 *Marking.* The ammeter dials shall be marked in accordance with the applicable drawing, with the markings, other than the numerals and major graduations, clearly defined in white on a black background. The numerals and the major graduations shall be finished with radio-active luminous compound conforming to grade 30M of Specification TT–R–58.

3.4.6.3 *Fastening.* The ammeter dials shall be secured to the body of the ammeters as shown on applicable drawings. The method of securing the dials shall be such that they will not become loose or turn when the ammeters are subjected to vibration.

3.4.7 *Pointers.* The ammeter pointers shall be suitably formed, light in weight, sufficiently rigid and damped to prevent oscillation under vibration, and shall be securely attached to the ammeter movements. The pointers shall be finished with radioactive, luminous compound conforming to grade 30M of Specification TT–R–58. The clearance between the pointers and the dials shall be the minimum practicable.

3.4.8 *Cover windows.* The cover windows shall be of clear, flat, unbreakable material, capable of retaining its transparency with age. The windows shall have a Mohs's scale minimum hardness of 2.5, and shall be free from flaws which might affect the readability of the dials.

3.5 Shunt.

3.5.1 *Rating.* The current rating of the shunt shall be as shown on applicable drawings (see 6.1).

3.5.2 *Components.*

3.5.2.1 *Leaves.* Shunt leaves shall be of suitable alloy having a negligible temperature coefficient of resistance. The leaves shall be securely assembled and soldered with silver solder (see 3.2.2) at each end. When specified, the assembled leaves shall be securely attached to a base with screws.

3.5.2.2. *Base.* Shunt base, unless shown otherwise on applicable drawings, shall be fabricated of phenolic material of a suitable type and grade conforming to Specification L–L–31.

3.5.3 *Attachments.* Unless otherwise specified, the assembled shunt shall be provided with the proper screws, nuts, and washers for attaching the shunt leads and circuit terminals. Provisions for mounting shall be as shown on the applicable drawings.

3.6 Performance.

3.6.1 *Calibration.* Each ammeter shall be so calibrated that, when held at a temperature of 80° ± 10°F., and with specified currents for corresponding dial readings applied, the angular variation of the centerline of the pointer from the centerline of the graduation will not exceed the limits specified on applicable drawings (see 6.1), as determined by the test specified in 4.3.4. The centerline of the pointer shall rest on the zero graduation when no current is flowing through the ammeter.

3.6.1.1 *Shunt calibration.* The shunt shall be so calibrated that, when held in an ambient air temperature of 80° ± 10°F., the resistance will be within the limits specified on the applicable drawing (see 4.3.4.2.1 and 6.1).

3.6.2 *Temperature influence.* Ammeters

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

shall function freely through a temperature range of —65° to 160°F. The readings at these two extremes of temperature shall vary from the room temperature readings by not more than ± 3 angular degrees, as determined by the test specified in 4.3.5.

3.6.3 *Overload capacity.*

3.6.3.1 *Ammeters.* Each ammeter shall withstand a 100-percent overload for a period of not less than 1 minute. The ammeter shall show no evidence of mechanical or thermal injury, and the calibration error shall be not more than 2 angular degrees as a result of the test specified in 4.3.6.1.

3.6.3.2 *Shunt.* The shunt shall withstand a temperature of 400°F. without evidence of thermal injury. The maximum operating temperature while carrying rated current shall exceed the ambient air temperature by not more than 300°F., as determined by the test specified in 4.3.6.2.

3.6.4 *Insulation resistance.* The insulation of the ammeters and of the shunt with bases shall withstand a potential of 440 volts r.m.s. at a frequency of 60 cycles per second for a period of not less than 10 seconds without evidence of cracking, charring, or other damage, as determined by the test specified in 4.3.7.

3.6.5 *Effect of vibration.* The ammeters and the shunt shall withstand vibration for a period of not less than 3 hours without evidence of disintegration of component materials or assemblies, as determined by the test specified in 4.3.8. The calibration error of the ammeter shall be not more than 2 angular degrees as a result of the specified test.

3.6.6 *Effect of shock.* The ammeters and the shunt shall withstand shocks caused by vehicular operation or from other causes without showing any tendency of the ammeter pointer to stick, delay to function, and evidence of disintegration of component materials or assemblies, as determined by the test specified in 4.3.9. The calibration error of the ammeter shall be not more than 2 angular degrees as a result of the specified test.

3.6.7 *Resistance to wear.* Ammeters shall meet the calibration requirements specified in 3.6.1 subsequent to the test specified in 4.3.10.

3.6.8 *Waterproofness.* The type I ammeter shall meet the waterproofness requirements specified in Specification MIL–E–13856 for type II, class 2 components (see 4.3.11).

3.6.9 *Resistance to corrosion.* The ammeters and the shunt shall show no evidence of corrosion when tested as specified in 4.3.12.

3.6.10 *Resistance to fungi.* The ammeters and the shunt shall show no harmful effects due to the growth of fungi when tested as specified in 4.3.13, and thereafter shall meet the requirements of 3.6.1.

3.6.11 *Dustproofness.* Type II ammeters shall operate satisfactorily in dust-laden atmospheres and after being tested as specified in 4.3.14 shall meet the requirements of 3.6.1.

3.7 **Protective finishes.**

3.7.1 *Chemical or metal finishes.* Unless otherwise specified, all exposed metal parts of the ammeters and the shunt not fabricated from corrosion-resistant materials shall be cadmium plated in accordance with type II, class A or B of Specification QQ–P–416, or equally treated to prevent corrosion.

3.7.2 *Painting.*

3.7.2.1 *Priming.* Ammeter bezels shall be given one coat of primer conforming to Specification TT–P–636. The prime coat shall have a film thickness of 1.0 to 1.5 mils.

3.7.2.2 *Finish coat.* Finish coat of the ammeter bezels shall be olive drab, semi-gloss enamel conforming to Specification TT–E–529, or type IV of Specification TT–E–485; or lacquer conforming to type I of Specification MIL–L–10287 and to spot 2315 of Specification TT–C–595. The finish coat shall have a film thickness of 0.75 to 1.25 mils.

3.8 **Marking.**

3.8.1 *Ammeters.* The manufacturer's name, initials, or trademark and the ordnance part number shall be permanently and

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

legibly printed or stamped on the ammeter dials.

3.8.2. *Shunt.* The Ordnance part number and the current rating shall be permanently and legibly stamped in a suitable location on the shunt.

3.8.3 *For Air Force use.* The physical item shall be marked for identification in accordance with MIL–STD–130.

3.9 **Workmanship.** Workmanship employed in fabricating the ammeters and the shunt shall be of a quality necessary to produce products free from all mechanical and electrical defects which might affect their functioning or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.

4.1.1 *Lot size.* Unless otherwise specified, a lot shall consist of all ammeters or shunt of one part number, from one manufacturer, submitted at one time for inspection.

4.1.2 *Sampling for inspection.* For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected in accordance with inspection level II of Standard MIL–STD–105.

4.1.3 *Sampling for tests.* From each lot which has passed the inspection specified in 4.2.3.2 the Government inspector shall select one each, or more, of the ammeters or shunt in each lot as samples for the specified acceptance tests (see 4.3.2.1). These samples may be taken from those already inspected as specified in 4.2.3.2.

4.2 Inspection.

4.2.1 *Place of inspection.* Unless otherwise specified (see 6.1), final inspection and tests for lot acceptance of the finished ammeters or shunt shall be made at the place of manufacture.

4.2.2 *Inspection facilities.* The contractor shall furnish for use by the Government inspector, such assistance, equipment, and facilities as the inspector may deem necessary in conducting the inspection and tests required herein.

4.2.3 *Visual, dimensional, and primary functional inspection.*

4.2.3.1 *For qualification.* Each specimen submitted for qualification tests shall be examined for appearance, finish, construction, completeness of manufacture and assembly, conformance to specified dimensions, dial and product markings, and workmanship.

4.2.3.2 *For acceptance.* The inspection sample, selected in accordance with 4.1.2, shall be inspected for conformance to the following quality levels on the bases of percent defective:

Classification of defects	*Acceptable quality level*
Major	1.0
Minor	2.5

4.2.3.2.1 *Classification of defects.* For inspection purposes, defects shall be classified as follows:

Major defects (1.0 AQL)

Calibration, including proper functioning and accuracy under normal conditions, in accordance with applicable drawings and with the general requirements of 3.6.1.

Minor defects (2.5 AQL)

Dimensions: Flanges, studs, threads, mounting holes, etc., as shown on drawings (see 3.3).

Markings: Legibility, permanance, luminosity of dials and other products (see 3.4.6.2 and 3.8).

Finish: Plated surfaces, painted surfaces, burrs, tool and die marks, etc. (see 3.7.1, 3.7.2 and 3.9).

Assembly: Completeness, tightness, etc. (see 3.9).

4.3 Tests.

4.3.1 *Qualification.* After inspection as specified in 4.2.3.1, specimens shall be subjected to the tests specified in table II in the sequence listed.

TABLE II. *Tests for qualification*

Ammeter sample	Sequence of tests	Shunt sample	Sequence of tests
1	4.3.4, 4.3.5, 4.3.7 4.3.10, 4.3.13	1	4.3.4, 4.3.6, 4.3.7
2	4.3.6, 4.3.12	2	4.3.8, 4.3.9, 4.3.12 4.3.13
3	4.3.8, 4.3.9, 4.3.11* of 4.3.14*		

* as applicable for particular type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

4.3.1.1 *Failure to Qualify.* Failure of any qualification tests specimen to pass any specified test shall be cause for refusal of the Government to conduct additional tests until it has been proven, to the satisfaction of the procuring activity that the faults revealed by the tests have been corrected.

4.3.2 *Acceptance and rejection.*

4.3.2.1 *Acceptance.* In addition to the inspection specified in 4.2.3.2, at least one ammeter from each lot shall be tested as specified in 4.3.11 and at least one shunt from each lot shall be tested as specified in 4.3.6.2, and subjected to such other tests specified herein as the Government inspector may deem necessary to determine conformance to this specification.

4.3.2.2 *Rejection.* Failure of any acceptance test sample to pass any specified test shall be cause for retest of twice the original number tested. Failure of any retest shall be cause for rejection of the lot.

4.3.3 *Test conditions.* The test under this specification shall be conducted in a laboratory free of dust, drafts, fumes, and products of combustion. Unless otherwise specified herein, room temperature shall be maintained at 80° ± 10° F., and relative humidity, as close as practicable, to 50 ± 5 percent. The ammeter shall be lightly tapped while being read.

4.3.4 *Calibration test.*

4.3.4.1 *Apparatus.* The testing apparatus shall include temperature-controlled air chambers capable of maintaining specified air temperatures within the range of 160° to —65° F., a laboratory-type dc millivoltmeter, a laboratory-type dc ammeter, a precision protractor, and sources of electrical energy.

4.3.4.2 *Procedure.* The ammeter shall be mounted in the air chamber, held at a temperature of 80° ± 10° F. in such a manner that the dial face is tilted backward at an angle of 30° from the vertical plane. The ammeter terminals shall be connected to a source of direct current which shall be adjusted to the values specified on applicable drawings as indicated on the laboratory ammeter. Readings shall be recorded by means of the precision protractor, at full-scale charge and full-scale discharge current, at zero current, and at intermediate charge and intermediate discharge current, to determine compliance with 3.6.1. Discharge readings are obtained by reversing the ammeter leads.

4.3.4.2.1 *Shunt.* The shunt shall be mounted in the air chamber held at a temperature of 80° ± 10° F. Direct current supply leads shall be connected to the heavy terminals through the ammeter and a controlling resistor. A current of not less than 10 percent of the full load rating of the shunt shall be passed through the shunt, and the voltage drop across the shunt shall be measured with the millivoltmeter at the small shunt terminals. The resistance shall be calculated by Ohm's law to determine compliance with 3.6.1.1.

4.3.5 *Temperature influence test.* The ammeter shall be tested as specified in 4.3.4, except that the ambient air and the ammeter shall be held at temperatures of —65° F. and 160° F. for sufficient lengths of time to insure complete stabilization of the ammeter at those temperatures but in no case less than 2 hours at each temperature. Readings shall be recorded at each temperature to determine compliance with 3.6.2.

4.3.6 *Overload test.*

4.3.6.1 Ammeter. The ammeter shall be tested as specified in 4.3.4, except that it shall be subjected to a current of twice the value indicated on applicable drawings for not less than 1 minute. At the conclusion of the test period the ammeter shall be examined for evidence of mechanical or thermal damage, and then shall be subjected to the test prescribed in 4.3.4, to determine compliance with 3.6.3.1.

4.3.6.2 *Shunt.* The shunt shall be subjected to sufficient current to cause the shunt leaf temperature to reach 400° F. The current shall then be disconnected and the shunt permitted to cool to room temperature. The shunt shall then be examined for evidence of thermal injury. The shunt shall be sub-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

jected to rated current for 5 minutes. At the conclusion of the 5-minute period, the temperature and voltage drop of the shunt shall be measured, to determine compliance with 3.6.3.2.

4.3.7 *Insulation resistance test.*

4.3.7.1 *Apparatus.* The apparatus for this test shall consist of a laboratory type ac voltmeter, and a source of alternating-current electrical energy.

4.3.7.2 *Procedure.* The specimen shall be subjected to 440 volts r.m.s., as indicated on the laboratory voltmeter, at a frequency of 60 cycles per second for 1 minute. When the test is being conducted on an ammeter, the current shall be applied between the case and one of the terminals. When the test is being conducted on the shunt, the shunt shall be placed on a flat metal plate, and the current shall be applied between the shunt and the plate. At the conclusion of the test, the insulation shall be examined for evidence of cracking, charring, or other damage, to determine compliance with 3.6.4.

4.3.8 *Vibration test.*

4.3.8.1 *Apparatus.* The apparatus for this test shall consist of a suitable device for mounting and vibrating the specimen through a total travel of 0.060 inch at 10 to 55 cycles per second.

4.3.8.2 *Procedure.* With the specimen rigidly mounted on the vibrating device, the unit shall be subjected to vibrations with an amplitude of 0.030 inch (0.060 inch total excursion) through a frequency range of 10 to 55 cycles per second. The unit shall be vibrated for 1 hour in the direction of each of the 5 major axes with continually varying frequencies being accomplished each minute. At the conclusion of the test, the unit shall be examined for evidence of damage, and then the ammeter shall be subjected to the calibration test specified in 4.3.4, to determine compliance with 3.6.5.

4.3.9 *Shock test.*

4.3.9.1 *Apparatus.* The apparatus for this test shall consist of a shock-testing machine similar to that specified in Specification JAN–S–44 with special fixtures as required for mounting the ammeter and shunt.

4.3.9.2 *Procedure.* With the specimen suitably attached to the movable carriage of the test machine, the carriage shall be raised to the height necessary to obtain a deceleration of 100 gravity units and then allowed to fall. The test shall be repeated 10 times. At the conclusion of the test, the unit shall be examined for evidence of damage, and then the ammeter shall be subjected to the calibration test specified in 4.3.4, to determine compliance with 3.6.6.

4.3.10 *Wear test.* Under the test conditions prescribed in 4.3.4, the ammeter shall be subjected to full-scale charge and full-scale discharge current for a total of 100,000 cycles. One cycle of operation shall consist of one complete oscillation of the pointer from extreme right to extreme left and return to extreme right, and shall be at the rate of 50 plus or minus 10 cycles per minute. At the conclusion of the cycling period, the ammeter shall be subjected to the calibration test specified in 4.3.4, to determine compliance with 3.6.7.

4.3.11 *Waterproofness test.* The type I ammeter shall be subjected to the waterproofness test specified in Specification MIL–E–13856 to determine compliance with 3.6.8.

4.3.12 *Corrosion-resistance test.* The specimen shall be subjected to the corrosion-resistance test specified in Specification QQ–M–151 for 200 hours. At the conclusion of the test the units shall be disassembled and examined for evidence of corrosion, to determine compliance with 3.6.9.

4.3.13 *Fungus-resistance test.* The specimen shall be subjected to the fungus-resistance test specified in Specification MIL–F–13927. At the time of each periodic performance test specified therein, units shall be tested in accordance with 4.3.4 to determine compliance with 3.6.10.

4.3.14 *Dustproofness test.* The type II ammeter shall be tested in accordance with Specification MIL–D–13570. After completion of the dust test the ammeter shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

tested as specified in 4.3.4 to determine compliance with 3.6.11.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* Ammeters and shunt shall be cleaned, if necessary, by any applicable process listed in Specification MIL–P–116 that will not damage the parts.

5.1.1.2 *Drying.* Ammeters and shunt shall then be dried in accordance with any drying procedure listed in Specification MIL–P–116 that will not damage the parts.

5.1.1.3 *Unit packaging.* Type I ammeters shall be packaged in accordance with Method A, type II ammeters in accordance with Method IIe, and shunts in accordance with Method III, of Specification MIL–P–116. Each ammeter, and each shunt, shall be individually wrapped and boxed, or otherwise protected in a manner which will effectively exclude dust, dirt, fumes, moisture, and other contaminants. Wrapping or packaging materials in direct contact with ammeter, or shunt, shall satisfy neutrality requirements of Specification MIL–B–121. Unless otherwise specified, each ammeter and each shunt shall be individually packaged in a container conforming to Specification PPP–B–566, MIL–B–4229, or PPP–B–676.

5.1.1.4 *Intermediate packaging.* Unless otherwise specified, intermediate package shall conform to Specification JAN–B–108, with 10 unit packages in each intermediate container. Only unit packages containing the same line item shall be placed in the intermediate container.

5.1.2 *Level C.* Ammeters or shunt shall be packaged in accordance with manufacturers standard practice provided that such will be sufficient to assure protection against deterioration and damage during shipment and subsequent interval preliminary to use.

5.2. Packing.

5.2.1 *Level A.* Ammeters or shunt packaged in accordance with 5.1.1.3 and 5.1.1.4, shall be packed in boxes in accordance with Specification JAN–P–106, MIL–B–107, MIL–B–138 or PPP–B–601 (overseas type), and in accordance with any additional instructions in the procurement documents.

5.2.2 *Level B.* Ammeters or shunt packaged in accordance with 5.1.1.3 and 5.1.1.4 shall be packed in boxes conforming to Specification NN–B–591, NN–B–621, NN–B–631, or PPP–B–601 (domestic type).

5.2.3 *Level C.* Ammeters or shunt shall be packed in exterior containers conforming to Consolidated Freight Classification Rules and Container Specifications for Rail Shipments or National Motor Freight Classification Rules and Container Specifications for Truck Shipments, as applicable.

5.3 Marking.

5.3.1 *Normal markings.* Unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of standard MIL–STD–129.

5.3.2 *Special markings.*

5.3.2.1 *Level A.* For overseas shipment of Ordnance material, the Ordnance flaming bomb symbol shall be stenciled on the same surface immediately adjacent to the overseas address, and the SNL group symbol (as ANL–G) shall be stenciled adjacent to each service color corner marking. The size of the symbols shall be commensurate with the letter size of the overseas address and the basic marking respectively.

5.3.2.2 *Level B.* For level B, the exterior container shall be conspicuously marked "LTD–MIL–PK".

5.3.2.3 *Level C.* For level C, the exterior container shall be conspicuously marked "MIN–MIL–PK".

5.3.2.4 *Precautionary marking.* The following precautionary marking shall appear conspicuously on two opposite sides of each interior package and each exterior container: "FRAGILE—DELICATE INSTRUMENTS —HANDLE WITH CARE".

6. NOTES

6.1 **Ordering data.** Procurement docu-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.

ments should specify the following:

(a) Title, number, and date of specification.

(b) Type (see 1.2).

(c) Applicable drawings (see 3.3 through 3.6.1.1).

(d) Level of preservation, packaging and packing required (see 5.1).

6.2 Qualification. In the procurement of products requiring qualification the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Commanding Officer, Detroit Arsenal, Center Line, Michigan.

Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Yards and Docks
Air Force

Other interests:
Army—ESigT
Navy—MCShS

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:39Z
Check the source to verify that this is the current version before use.